MILITARY SPECIFICATION

PROJECTILE, 105MM, SMOKE, WP,

M60E1; FILLING ASSEMBLY

1. SCOPE

1.1 This specification covers the filling assembly with WP, of the 105MM, M60E1 projectile without fuzing components.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-215 - Chemical Agent, White Phosphorus (WP).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

US ARMY MUNITIONS COMMAND

F9225939 - Projectile, 105MM; Smoke, WP, M60E1, Filling Assembly.
DL8837835 - Pallet for Projectile Metal Parts.
DL8861520 - Marking Diagram for Pallets for Projectile 75MM to 120MM.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 *Materials.* All materials cited on Drawing F9225939 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the projectile shall conform to the specifications and drawings listed on Drawing F9225939 and subsidiary drawings.

3.2 Filling and assembly.

3.2.1 Filling. The projectile, clean and free of foreign matter, shall be filled with molten WP conforming to MIL-C-215 (see 6.2 and 6.3). When water is in contact with the WP during the filling operation, the water shall be buffered by a suitable alkaline salt to maintain a pH value between 6.5 and 8.0. After test for leakage, the WP shall be solidified with the projectile in a vertical nose-up position.

3.2.2 Filling height. There shall be a 7 to 12 percent void, by total volume, existing above the surface of the water over the WP in the filled and closed projectile. The depth of water remaining in the filled and closed projectile shall not exceed 1/4 inch. The projectile shall be radiographed in a vertical, nose-up position while the temperature of the WP is 130 ± 10°F when tested in accordance with 4.2.4.1 to determine filling *height and seating of the burster casing (see 6.3).*

3.2.3 Assembly. The projectile shall be assembled in accordance with Drawing F9225939 except that the assembly shall not include the following: burster charge assembly, fuze well cup and pad.

3.3 Cleaning after closing. The exterior surface of the filled and closed projectile, fuze threads and the interior of the burster casing shall be cleaned thoroughly to remove all grease, oil, and WP contamination. (See 6.4)

3.4 Leakage. When tested in accordance with 4.2.4.2, the closed and cleaned projectile shall not leak when maintained at a temperature of 160 ± 5°F for a period of 8 hours. Alternate method: The projectile shall not leak when maintained at a temperature of 207 to 212°F for a minimum period of 15 minutes.

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3.5 Workmanship. The projectile shall be free of all foreign matter (oil, grease, contamination) and damage such as dents, cracks, and deformation which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of projectiles, of one interfix number, from one manufacturer, filled by the same filling installation, under essentially the same conditions. However, no more than one interfix lot of burster casings from one manufacturer shall be represented in any one interfix lot of filled projectiles.

4.2.2 Sampling. Sampling for examination and nondestructive tests shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination and nondestructive test. Sample projectiles shallbe examined and tested in accordance with the classification of defects and MIL-STD-105.

4.2.3.2 For critical defects. Each projectile shall be inspected for the critical characteristics listed in the classification of defects.

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4.2.3.3 Classification of defects.

(a) Projectile, 105MM: smoke, WP, M60E1, filling assembly (Dwg F9225939).

Categories	Defects	Acceptance standards
Critical:		
1	Leakage	4.2.4.2
2	WP contamination	
Major:	AQL 1.0 percent defective	
101	Burster casing not seated	4.2.4.1
102	Threads damaged or contaminated	
103	Bourrelet diameter (maximum) after protective finish	
104	Filling excessive	4.2.4.1
105	Filling insufficient	4.2.4.1
106	Excess water	4.2.4.1
107	Marking incorrect	
108	Protective finish missing	
109	Workmanship (3.5)	
Minor:	AQL 2.5 percent defective	
201	Rotating band damaged	
202	Marking missing or illegible	
203	Threads not coated	

(b) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Pallets incorrect
102	Protective sleeves missing
103	Strapping loose or missing
104	Marking illegible or incorrect

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4.2.4 Tests.

4.2.4.1 Filling height. The projectile shall be radiographed while the WP is molten as specified, and the projectile stored in a vertical nose-up position.

4.2.4.2 Leakage. The projectile shall be placed in an oven in a vertical, filling-end-down position (for the alternate method filling-end-up position) and maintained at the specified temperature for the required period of time. When testing at 207 to 212°F, the projectile shall be measured by thermocouples inserted in the burster wells of at least two projectiles located in the center of each oven bay batch. The emission of smoke constitutes leakage.

5. PREPARATION FOR DELIVERY

5.1 Packing, level A. The filled and closed projectile shall be packed in accordance with Drawing F8837835.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Drawing C8861520.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.2 Caution. Because of the hazardous nature of WP operations, filling installations are advised to take all necessary precautions required by proper industrial hygiene and safety practices to protect personnel from injury while filling projectiles in accordance with this specification.

6.3 Filling aid. Projectiles filled with approximately 3.9 ± 0.3 pounds of WP have been found to comply with 3.2.2. This information is furnished as an aid to procurement and not as a criterion for filling. The void space above the water over the WP may be either air or inert gas.

6.4 Decontamination. The projectiles have been decontaminated by hot water and detergent in combination with steam lances and wire brushes. This information is furnished as an aid to production and not as a filling line design criteria.

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6.5 Supersession data. This specification supersedes Edgewood Arsenal Purchase Description 196-131-892(1), dated 13 August 1968.

Custodian:

Army - MU

User activities:

Navy - OS
Air Force - 70

Preparing activity:

Army - MU(EA)

Project No. 1315-A751

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